

Response of Indian Iron and Steel Companies Towards Biodiversity: An LLM-Based Assessment of Corporate Biodiversity Disclosures and Alignment with the Global Biodiversity Framework

Sagnik Roy¹, Mousami Prasad^{2*}

¹Indian Statistical Institute, Kolkata

²Indian Institute of Technology Kanpur

Abstract

Iron and steel production depends heavily on natural resources like water availability, ecosystem services, and waste assimilation capacities. At the same time, steel manufacturing significantly harms biodiversity through resource extraction, land and water pollution, emission of GHG as well as non-GHG air pollutants, habitat degradation, and the release of contaminants into soil and water. Growing concerns over biodiversity loss have increased stakeholder demands for greater transparency regarding the environmental impacts and dependencies of industrial operations, leading companies to disclose ESG (Environmental, Social, Governance) objectives into their reporting practices.

Despite the growing prevalence of biodiversity-related disclosures, it remains unclear whether these disclosures address the biodiversity issues most material to what the company does and whether they translate into actionable outcomes. Existing assessments of biodiversity reporting remain limited and difficult to scale.

Here, we develop a Large Language Model (LLM)-based framework to evaluate biodiversity disclosures in the Indian iron and steel sector. We analyze the 2024-25 annual and sustainability reports of 100 iron and steel companies automatically extracting biodiversity-related statements and classifying them as goals, commitments, or SMART - Specific, Measurable, Achievable, Relevant, and Time-bounded targets. Statements are subsequently mapped to the 23 targets of the Global Biodiversity Framework and assessed against sector specific biodiversity impacts and dependencies identified through the ENCORE (Exploring Natural Capital Opportunities, Risks and Exposure) framework.

We find that these companies are addressing biodiversity issues in which government regulations are put in place. 85% of the statements are about pollution reduction and disclosures related to biodiversity, which are already regulated by the government. However, less than 20% of the statements talked about protection of genetic diversity and halting species extinction and conservation of land and water which is one of the major impacts of iron and steel industries. Moreover, most disclosures take the form of broad goals and commitments rather than fully developed SMART targets which account only for 5% of total statements. These findings suggest that the principal challenge in corporate biodiversity reporting is in tackling issues which are not regulated by the government where they lack on taking any actions on those issues. Thus, it brings out the necessity of regulations and accountability frameworks to motivate companies to act on these issues.

*Corresponding author:

1 Introduction

According to a report by Centre for Science and Environment(CSE) in 2012, the Indian iron and steel sector is one of the poorest performing industries in terms of environmental performance, being highly non-transparent and poor in disclosure of environmental impacts. Under the Green Rating Project, a sustainability metric designed by CSE, the sector received a "One Leaf" rating, and the best performing company achieved a rating of "Three Leaves",which indicates average performance.

In recent years, growing concerns over biodiversity loss, climate change, and ecosystem degradation have forced companies to disclose their environmental impacts. Investors, regulators, and other stakeholders now increasingly require companies to disclose environmental, social, and governance (ESG) objectives and demonstrate their commitment to sustainable practices through transparent reporting. However, the extent to which biodiversity related disclosures in the iron and steel sector translate into meaningful and measurable conservation actions remains poorly understood.

In this study, we analyse the latest annual reports (2024–25) of 100 iron and steel companies operating in India to evaluate their approach towards biodiversity conservation and restoration. We develop a Large Language Model (LLM)-based framework that automatically extracts biodiversity-related statements from corporate reports. The reports are first processed through a text cleaning pipeline to remove irrelevant textual elements. The cleaned text is then analysed using an LLM to identify statements pertaining to biodiversity conservation. These statements are subsequently classified as goals, commitments, or SMART targets. The extracted information is stored in a structured JSON format and further analysed using multiple evaluation metrics to assess the quality and extent of corporate biodiversity disclosures.

2 Methodology

This study develops a Large Language Model (LLM)-based framework to evaluate biodiversity disclosures in the annual reports of Indian iron and steel companies.

2.1 Dataset

2.1.1 Company Selection

A list of 100 iron and steel companies have been selected for the study whose annual reports are publicly available through BSE(Bombay Stock Exchange), NSE(National Stock Exchange) or in their corresponding websites. To ensure the study was not confounded by market capitalizations, firms ranging from small cap to large cap were chosen. The study analyzes reports for the 2024–25 financial year, representing the most recent reporting period with publicly available annual reports.

2.1.2 Report Preprocessing

Each annual report is processed page by page and the extracted text from each page is saved as an individual element in a list. The pages are preprocessed to remove null characters, redundant whitespaces and repeated punctuation to reduce noise and enable cleaner extraction by the LLM.

2.2 Biodiversity Assessment Framework

2.2.1 Global Biodiversity Framework (GBF)

Biodiversity is defined as the extraordinary variety of life found in a place. It plays an essential role in keeping the ecosystem stable and is fundamental for human well-being. The Kunming-

Montreal GBF establishes a global roadmap comprising 23 targets aimed at halting and reversing biodiversity loss by 2030. The targets address different issues such as reduction of biodiversity threats, integration of urban life with nature and developing tools and solutions for effective implementation. For example, Target 7 tells to reduce pollution levels that are not harmful to the biodiversity. Owing to its comprehensive coverage of global biodiversity priorities, the GBF serves as the reference framework for evaluating corporate biodiversity disclosures in this study. Each statement extracted by the LLM is again mapped to one of the GBF targets depending upon which specific area it is targeting.

2.2.2 ENCORE Framework

ENCORE (Exploring Natural Capital Opportunities, Risks and Exposure) is a tool that sets out how the economy depends and impacts on nature. Industries can use ENCORE to identify nature related risks they are exposed to through their operations. ENCORE is designed to be a useful entry point to nature-related assessment initiatives, such as the new risk management and disclosure framework developed by the Taskforce on Nature-related Financial Disclosures (TNFD). It lists out the impacts & dependencies for each sector of the industry along with their materiality ranging from low to high. We use the ENCORE impacts and dependencies to select the GBF targets concerning the iron and steel sector and weighting them by their materiality.

2.3 LLM-Based Biodiversity Statement Extraction

The **Gemini 3.1 Flash-Lite** large language model was used for all statement extraction and classification tasks. The extraction prompt instructed the model to extract statements only related to biodiversity conservation, ecological restoration and biodiversity management using certain keywords. In addition to that, negative prompts were added to explicitly exclude statements unrelated to biodiversity, such as those concerning general corporate governance, social responsibility or general governance. The extracted statements were stored in a structured JSON format, to be saved for later classification and analysis.

2.4 Statement Classification

After extraction of the statements, the LLM now classifies each statement into the following 3 categories:

- **Goals** - Goals are vague and broad aspirational statements lacking metrics and timelines
- **Commitments** - Commitments are specific intentions that usually lack measurability and deadlines
- **SMART Targets** - SMART Targets are statements which are specific, measurable, achievable, relevant and time-bounded.

2.5 Mapping Statements to GBF Targets

As described previously, the ENCORE framework was used to identify the biodiversity impacts and dependencies most material to the iron and steel sector. These impacts and dependencies were subsequently mapped to the corresponding Global Biodiversity Framework (GBF) targets, with each target assigned a materiality weight based on its relevance to the sector. Each statement extracted from the annual reports are mapped to one or more GBF targets based on their semantic content. Since a single statement could address differently biodiversity issues, it could be associated with different GBF targets.

2.6 Quantitative Analysis

2.6.1 Disclosure Composition

Following the extraction, classification, and mapping of biodiversity-related statements, we perform some analyses to characterize the biodiversity disclosure patterns of each company. First, the proportions of goals, commitments, and SMART targets were computed to assess the quality and maturity of corporate biodiversity disclosures. A higher proportion of SMART targets would indicate that a company's biodiversity initiatives are more specific and actionable, whereas a larger proportion of goals and commitments reflects broader intentions with limited accountability.

Next, the distribution of disclosure types across the Global Biodiversity Framework (GBF) targets was analysed to determine whether companies are setting measurable targets for the biodiversity issues that are most material to their operations. By combining the disclosure classifications with the ENCORE-derived materiality scores, we evaluate whether high-priority biodiversity issues are primarily addressed through goals, commitments, or SMART targets.

In addition, the frequency with which each GBF target is addressed across all companies was computed to identify the biodiversity issues receiving the greatest corporate attention. Conversely, GBF targets that were absent or infrequently addressed were identified together with their associated materiality scores, enabling the identification of critical gaps in biodiversity reporting. This analysis provides insights into whether companies are focusing predominantly on governance-oriented sustainability initiatives or are actively addressing the biodiversity issues most relevant to the iron and steel sector.

Finally, to facilitate an objective comparison of biodiversity disclosure performance across companies, we propose a Biodiversity Disclosure Index (BDI) that combines the quality of disclosed statements with the materiality of the biodiversity issues they address.

2.7 Biodiversity Disclosure Index

Let $\mathcal{G}$ denote the set of GBF targets identified by ENCORE. For each GBF target $i \in \mathcal{G}$, let

$$\begin{aligned} I_G(i) &= \begin{cases} 1, & \text{if target } i \text{ is addressed through a Goal,} \\ 0, & \text{otherwise,} \end{cases} \\ I_C(i) &= \begin{cases} 1, & \text{if target } i \text{ is addressed through a Commitment,} \\ 0, & \text{otherwise,} \end{cases} \\ I_T(i) &= \begin{cases} 1, & \text{if target } i \text{ is addressed through a SMART Target,} \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

The weights assigned to Goals, Commitments, and SMART Targets are

$$w_G = 1, \quad w_C = 2, \quad w_T = 3.$$

Further, let

$$M_i \in \{1, 2, 3, 4, 5\}$$

denote the ENCORE-derived materiality weight associated with GBF target i , where 1 corresponds to very low materiality and 5 corresponds to very high materiality.

The Biodiversity Disclosure Index (BDI) is then defined as

$$\text{BDI} = \frac{\sum_{i \in \mathcal{G}} M_i (w_G I_G(i) + w_C I_C(i) + w_T I_T(i))}{6 \sum_{i \in \mathcal{G}} M_i}.$$

We normalize the index by $6 \sum_i M_i$ as it can be the highest score achieved (when each target is addressed by all types of statements).

The index thus defined assigns higher scores to companies that state more measurable SMART targets addressing biodiversity issues of high materiality, while comparatively lower scores are assigned to companies whose disclosures consist primarily of broad goals or commitments concerning less material biodiversity issues. Consequently, the normalised score provides a measure of meaningful comparisons across companies.

3 Results

3.1 Disclosure Classification

Following the extraction and classification of biodiversity-related statements, we analysed the overall composition of biodiversity disclosures across the 100 iron and steel companies included in this study. 9 companies had no biodiversity related statements extracted from their annual reports, indicating an absence of explicit biodiversity disclosures. Among the remaining companies, the number of extracted statements varied considerably, ranging from 1 to 269 statements, with an average of 27 statements per company. This large variation indicates that companies do not have any fixed outline for disclosure and there are substantial differences in their biodiversity related initiatives.

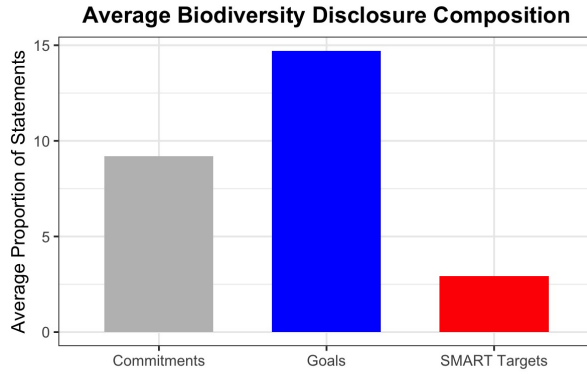


Figure 1: Distribution of Type of Statements

Overall, biodiversity reporting was dominated by broad and aspirational statements rather than actionable and time bound targets. On an average, only **5%** of the extracted statements were **SMART** targets, while approximately **55%** were classified as goals and **33%** as commitments. This indicates that companies are acknowledging biodiversity related issues, as qualitative goals rather than quantitative and achievable goals.

Among all companies, Tata Steel reported the highest number of SMART targets, with 46 SMART targets out of a total of 185 biodiversity-related statements, followed by JSW Steel and SAIL. On the other hand, companies such as Aditya Ultra Steels and Remi Edelstahl reported very few statements and had 0 SMART targets.

3.2 Alignment with Material GBF Targets

As discussed earlier, the ENCORE framework was used to identify the Global Biodiversity Framework (GBF) targets that are most material to the iron and steel sector. Based on the sector specific impacts and dependencies, 11 GBF targets (Targets 1, 2, 3, 4, 7, 8, 11, 12, 14, 15, and 16) were identified as priorities. These targets primarily focus on reducing threats to biodiversity through the conservation of species and genetic diversity, conservation of land,

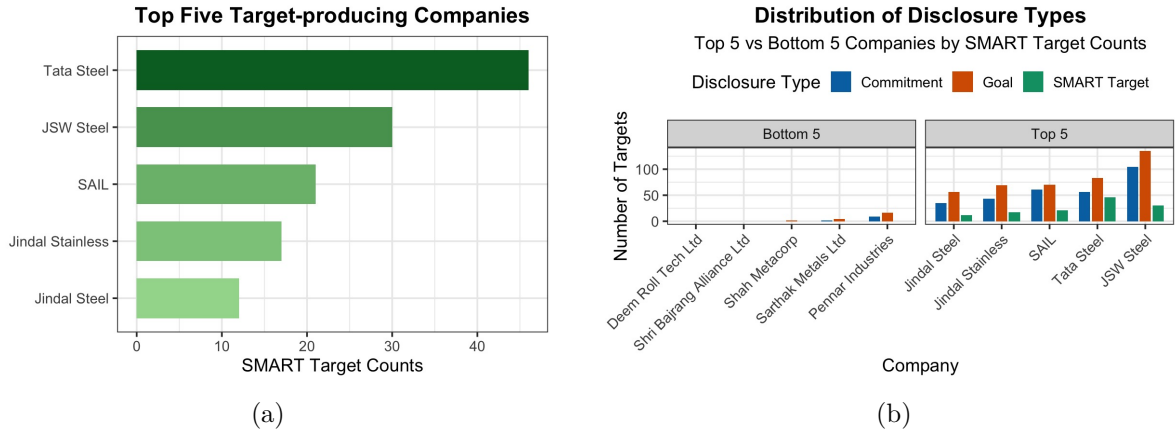


Figure 2: Distribution of Targets in Top 5 and Bottom 5 companies

water and seas, restoring degraded ecosystems, reducing pollution, and integrating biodiversity considerations into corporate disclosures.

To evaluate whether companies are addressing the biodiversity issues most relevant to their operations, we compared the selected GBF targets with those represented in each company’s biodiversity disclosures. This indicates whether companies are aware about their impact on biodiversity and thus take measures accordingly.

Across the study, companies failed to address a substantial proportion of the material GBF targets. On average, approximately 6 of the 11 material targets were unaddressed, showing a considerable gap between expected and reported biodiversity priorities. Only a small number of companies, namely Tata Steel, Jindal Stainless, JSW Steel, and APL Apollo Tubes, addressed all the targets in their annual reports. In contrast, companies such as Riddhi Steel and Tubes Ltd., Shri Bajrang Alliance Ltd., and Remi Edelstahl failed to address any of the material GBF targets, indicating a complete absence of relevant biodiversity disclosures.

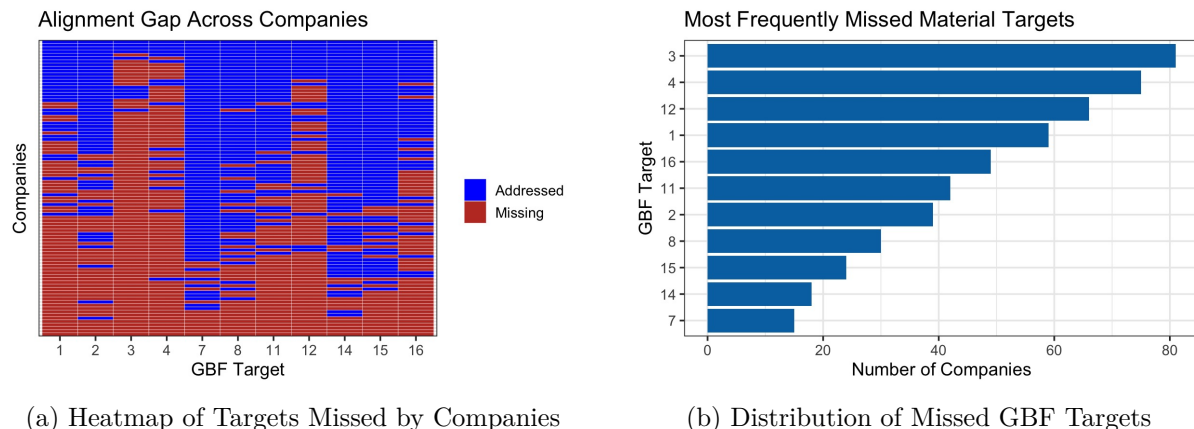
GBF Target	No of Companies
3	81
4	75
12	66
1	59
16	49
11	42
2	39
8	30
15	24
14	18
7	15

Table 1: Number of companies missing each target

From Table 1 it is evident that GBF targets 3 and 4, which relate to the restoration of degraded ecosystems and the conservation of species, were absent from the disclosures of nearly 80% of the companies despite being high materiality targets. In contrast, GBF Targets 7 and 15, which focus on pollution reduction and the integration of biodiversity considerations into corporate policies and governance, were disclosed much more frequently.

The high level of reporting on pollution control is encouraging, particularly because pollution is one of the high material impacts of iron and steel production. However, it is important to

note that pollution control in India is enforced by an extensive legal framework, including the **Environment (Protection) Act, 1986**, the **Air (Prevention and Control of Pollution) Act, 1981**, and the **Water (Prevention and Control of Pollution) Act, 1974**, which are implemented and monitored by the **Central Pollution Control Board (CPCB)**. On the other hand, ecosystem restoration and species conservation are subject to considerably fewer sector specific reporting requirements. The poor reporting of GBF targets 3 and 4 therefore suggests that, in the absence of strict regulations, companies usually allocate fewer resources to biodiversity conservation measures.



Next we examined how the high materiality biodiversity targets are being addressed by companies. Specifically, we calculated the proportions of goals, commitments, and SMART targets for each of the six GBF targets (Targets 3, 4, 7, 11, 12, and 14) which have a high materiality rating

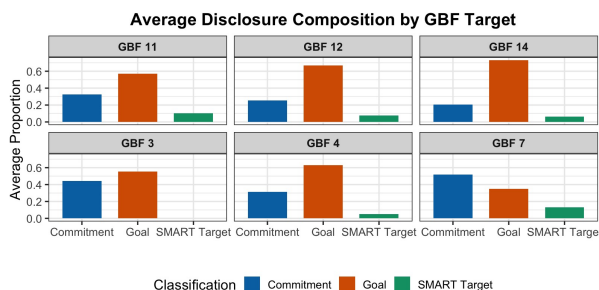


Figure 4: Distribution of Statements among GBF Targets

Figure 4 shows that, with the exception of GBF Target 7, the majority of disclosures are classified as goals rather than commitments or SMART targets. SMART targets constitute less than 20% of the statements across the GBF targets. This indicates that while companies are increasingly acknowledging the biodiversity issues they need to address, these disclosures are aspirational and vague and are rarely measurable or achievable.

3.3 Correlation with Market Capitalization and Net Profit Margin

The Biodiversity Disclosure Index (BDI), introduced in the previous section, provides a quantitative measure of the quality and relevance of a company's biodiversity disclosures. To investigate whether biodiversity reporting is associated with company performance and size, we examined its relationship with market capitalization and net profit margin (NPM) for the financial year 2024–25.

Companies for which market capitalization or net profit margin data were unavailable were excluded from the analysis. Since both variables were highly varied and contained several

outliers, Spearman’s rank correlation coefficient was used. Spearman’s correlation is a non-parametric measure that is robust to outliers and takes into consideration only the ranks of the variables. Owing to the relatively large sample size ($n > 10$), p -value was evaluated using the asymptotic distribution of the correlation coefficient.

A strong positive correlation was observed between the Biodiversity Disclosure Index and market capitalization ($\rho = 0.85$, $p < 2.2 * 10^{-16}$), indicating that larger companies generally exhibit more higher quality biodiversity disclosures.

A moderate positive correlation was also observed between the Biodiversity Disclosure Index and net profit margin ($\rho = 0.42$, $p < 5.6 * 10^{-05}$). The analysis included one notable outlier, Oil Country Tubular Ltd., which reported a net profit margin of -386.25% during FY 2024–25. Despite the outlier, the positive association remained statistically significant, suggesting that companies with stronger financial performance generally tend to report more mature biodiversity disclosures.

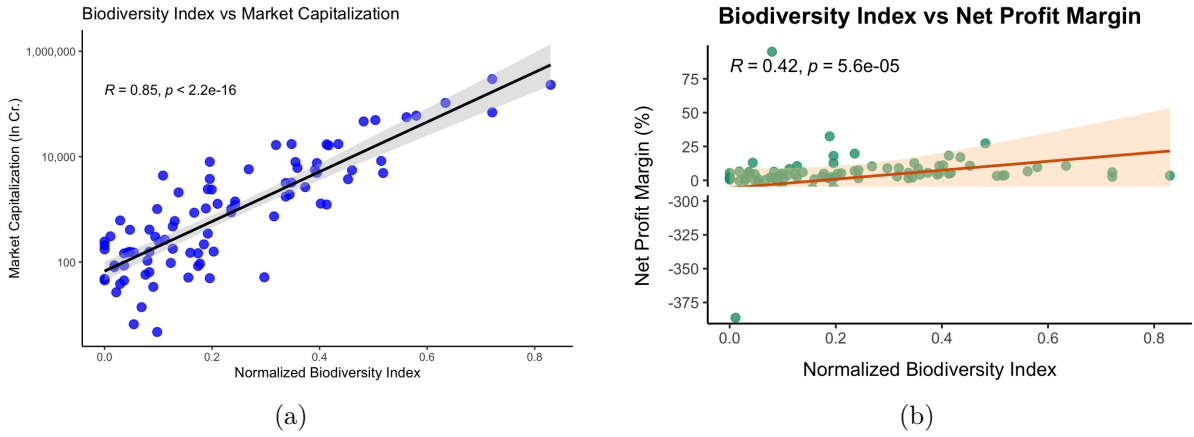


Figure 5: Scatter-plots of BDI with Market Cap & Net Profit Margin of 2024–25

Overall, these findings indicate that biodiversity disclosure quality is positively associated with both company size and financial performance. Larger and more profitable firms are more likely to invest in comprehensive biodiversity reporting and develop measurable conservation targets, whereas smaller or financially constrained companies typically disclose fewer biodiversity initiatives and rely more heavily on broad goals and commitments. These results highlight the need for improved reporting guidance and targeted regulatory support to encourage high-quality biodiversity disclosures across companies of all sizes, rather than limiting robust reporting practices to large, well-resourced organizations.

4 Conclusion

The results of this study show that the Indian iron and steel sector still has considerable scope for improving its biodiversity disclosures. Only a small proportion of biodiversity related statements qualify as SMART targets, with most disclosures being goals or commitments that lack measurability and clarity. Companies also tend to focus on biodiversity issues that are already regulated, such as pollution control, while high materiality issues like ecosystem restoration and species conservation remain largely unattended in their reports.

Our analysis also indicates that larger companies generally have better biodiversity disclosures than smaller companies. This suggests that company size and the availability of financial resources may influence the ability to invest in sustainability initiatives and prepare comprehensive biodiversity reports. Smaller companies may lack the resources required to adopt sustainable technologies or implement effective biodiversity management practices.

These findings highlight the need for stronger regulations and reporting standards that encourage companies to set measurable and time bound targets, particularly for high-materiality issues. At the same time, providing financial and technical support to smaller companies could help improve biodiversity management across the sector. Together, these measures can strengthen corporate accountability and contribute towards achieving the objectives of the Global Biodiversity Framework.

References

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